

Vishwas Kothari

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Graduating May 2027 | Available June 2027

SUMMARY

M.S. Computer Science candidate at CU Boulder building machine-learning and retrieval systems in Python. Combines statistical evaluation with tested code, reproducible workflows and interpretable outputs. Projects span financial AI and credit modeling, alongside numerical optimization and satellite-data research at ISRO.

EDUCATION

University of Colorado Boulder

Aug 2025 – May 2027

Professional Master's in Computer Science, **GPA: 3.74/4.0**

Boulder, CO

Coursework: Design & Analysis of Algorithms, Database Systems, Neural Networks & Deep Learning.

Database coursework: Processed one million records with PySpark and Spark SQL; cut repeat-query latency from **176 ms to 20 ms** using Redis in a Node.js service over a 500,000-row PostgreSQL table.

Vellore Institute of Technology

Aug 2021 – Jun 2025

B.Tech in Information Technology, GPA: 8.49/10.0

Vellore, India

EXPERIENCE

Space Applications Centre, Indian Space Research Organisation (ISRO)

Ahmedabad, India

Research Intern, Biological Oceanography Division

Jan 2025 – Jul 2025

- **Resolved convergence** failures across three orders of magnitude using log-space reparameterization in a Levenberg–Marquardt bio-optical model; implemented Particle Swarm Optimization for comparison.
- **Validated estimates** against in-situ observations, reaching **$R^2=0.987$** on reflectance reconstruction and log-space **$R^2=0.879$** across 105 chlorophyll samples; assessed agreement using RMSE, MAPD, relative bias and R^2 .
- **Automated colocation** of OCM and MODIS Aqua satellite data in Python/Linux during the internship's second phase; produced six visualization types and a 44-page report presented across research divisions.

SELECTED PROJECTS

FinRAG Auditor | Hybrid Retrieval and Validated Tool Execution

GitHub

- **Improved retrieval** by **10.38 mean reciprocal rank points** over TinyBERT (95% paired bootstrap CI: 8.27–12.61) by selecting a BGE reranker for hybrid BM25/dense search across 883 development questions and 8,783 chunks.
- **Validated execution** with **114 tests** at approximately 93% combined line/branch coverage for a bounded LangGraph workflow with citation/operand checks, restricted Decimal arithmetic and abstention; added Ruff, mypy and Docker build/CLI smoke tests in CI.

FinAgent-Eval | Document AI and LLM Evaluation

GitHub

- **Benchmarked models** on 397 questions across 79 financial-document images through a five-model pipeline with unit-aware scoring, bootstrap confidence intervals, a 13-label failure taxonomy and a seven-tab Streamlit dashboard.
- **Corrected scoring** for 35 of 151 previously flagged failures by repairing answer normalization, including financial units; regenerated versioned metrics and expanded validation to **96 tests** with CI checks for artifact and metric drift.

XAI Credit Lens | Credit-Default Modeling and Explainability

GitHub

- **Modeled default** across 30,000 historical credit-card records, reaching **0.782 test ROC AUC** and a **0.180 Brier score** by comparing Random Forest, XGBoost and LightGBM with Optuna and stratified five-fold cross-validation.
- **Audited disparities** across four demographic attributes; separately developed a Streamlit prototype with SHAP explanations and what-if queries, plus model-risk documentation templates for reviewing predictions and limitations.

Implicit Bias of Adam vs. SGD | Optimization and Robustness

GitHub

- **Evaluated optimizer bias** against numerically solved max-margin references using CVXPY across five seeds, observing mean cosine similarities of 0.990 for gradient descent (L2) and 0.988 for Adam (L-infinity).
- **Measured robustness** in a five-seed binary-MNIST CNN experiment: mean out-of-distribution accuracy dropped by 30.2 percentage points for SGD versus 13.6 for Adam when a spurious correlation flipped from 99% to 1%, despite near-identical in-distribution accuracy.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, R, MATLAB

Machine Learning: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM, Optuna

AI and Retrieval: RAG, LangGraph, LangChain, BM25, dense embeddings, reranking, LoRA/PEFT

Statistics and Explainability: Cross-validation, bootstrap intervals, calibration assessment, SHAP, LIME

Data Systems: PostgreSQL, Redis, Apache Spark, PySpark, Spark SQL, Pandas, NumPy, Linux

Engineering: Git, Docker, GitHub Actions (CI), pytest, Ruff, mypy, Streamlit

PUBLICATION, PEER REVIEW AND CERTIFICATION

Publication: Kothari, V. and Krishwanth, B. (2025). *Empowering Survivors: Ethical Artificial Intelligence for Countering Violence Against Women (VAW)*. Springer.

Peer Reviewer: *Public Health; Social Sciences & Humanities Open* (Elsevier, 2026–Present).

Certification: AWS Certified Cloud Practitioner (2024).